

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

NAME: SARANYA.R.A

REG NO: 192572052

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

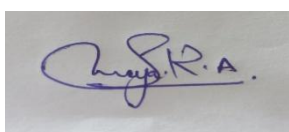
Analytical Problem Solving

Code of Conduct:

I Saranya.R.A(192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

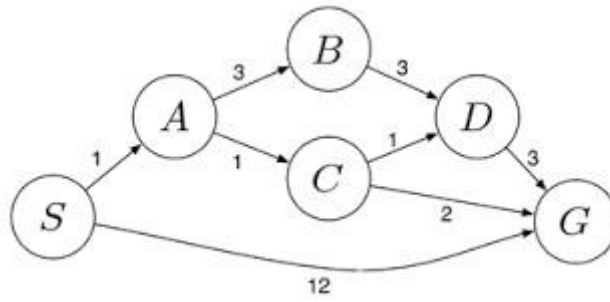


Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (48)
Problem Understanding	20% (9.6)	/2	/2	/2	/2	/2	/10
Method Selection	20% (9.6)	/2	/2	/2	/2	/2	/10
Solution Process	25% (12)	/2.5	/2.5	/2.5	/2.5	/2	/12
Accuracy of Calculation	20% (9.6)	/2	/2	/2	/2	/1	/9
Interpretation of Results	15% (7.2)	/1.5	/1.5	/1.5	/1.5	/1	/7
Total per Question	48 Marks	/10	/10	/10	/10	/8	/48

ANSWER:

1. Given:

- 4-gallon jug = A
- 3-gallon jug = B
- Initial state = (0,0)
- Goal = (2,0)

Solution:

Step	Operation	State (A,B)
1	Fill 3-gallon jug	(0,3)
2	Pour 3-gallon into 4-gallon	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until it is full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour 2 gallons into 4-gallon jug	(2,0)

Thus, exactly 2 gallons are obtained in the 4-gallon jug.

2.i. Percepts

The rover receives information through cameras and sensors, such as terrain, rocks, obstacles, temperature, location, and soil composition.

ii. Operating Environment

The environment is partially observable, stochastic, dynamic, sequential, continuous, and unknown.

iii. Actions

The rover can:

- Move and turn
- Avoid obstacles
- Capture images
- Collect and analyze samples
- Select exploration locations
- Send information to Earth

iv. Performance Evaluation

The rover can be evaluated based on:

- Scientific data collected
- Area explored
- Samples analyzed
- Energy efficiency
- Safe navigation
- Successful completion of mission goals

v. Suitable Agent Architecture

A model-based, goal-based, and utility-based agent is most suitable. It can maintain information about its surroundings, select goals, and choose the best action under uncertain conditions.

3. 8-Queens Problem

The objective is to place 8 queens on an 8×8 chessboard so that no two queens attack each other.

Problem Components

- Initial State: Empty chessboard.
- State: A partial arrangement of queens.
- Actions: Place a queen in a safe position.
- Transition: Add a queen to the next row.
- Goal Test: All 8 queens are placed with no row, column, or diagonal conflicts.
- Path Cost: Each queen placement can have a cost of 1.

Suitable Search Strategy

DFS with Backtracking is suitable. If a queen cannot be placed safely, the algorithm goes back to the previous position and tries another choice.

One valid solution is:

[1, 5, 8, 6, 3, 7, 2, 4]

This represents the column position of the queen in each row.

4. OLA Cab Booking Agent

A goal-based agent can be used for this problem. The goal is to select a suitable cab and reach the destination.

Pseudocode

START

Enter pickup location

Enter destination

Find available cabs

IF no cab is available

 Display "No cabs available"

ELSE

 Display cab types

 Select preferred cab

 Show estimated fare and time

 IF customer confirms

 Book cab

 Assign driver

 Navigate to destination

 Display "Trip completed"

 ELSE

 Cancel booking

END IF

END IF

STOP

The agent can also use factors such as cost, comfort, travel time, and availability to recommend the best cab type.

5. Uniform Cost Search (UCS)

The objective is to find the least-cost path from S to G.

UCS Expansion

- Start at S
- $S \rightarrow A = 1$
- $A \rightarrow C = 1$, total = 2
- $C \rightarrow D = 1$, total = 3
- $C \rightarrow G = 2$, total = 4

Other possible paths:

- $S \rightarrow G = 12$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 6$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = 10$

Therefore, the optimal path is:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Minimum Cost

$$1 + 1 + 2 = \boxed{4}$$

Hence, using Uniform Cost Search, the logistics company should choose $S \rightarrow A \rightarrow C \rightarrow G$, with a minimum total cost of 4.